

6**Applications of ANN****6.1 : Applications of ANN in Symbol Recognition**

Q.1 What is the central concept of pattern recognition ? How ANN plays role in pattern recognition ? [SPPU : 5 Marks]

- Ans. :** • Neural networks is a best choice for pattern classification due to its ability of a multilayer feedforward neural network to form complex decision regions in the pattern space for classification.
- Many pattern recognition problems, especially character or other symbol recognition and vowel recognition, have been implemented using a multilayer neural network. It should be well noted that multilayer neural networks are not directly applicable for situations where the patterns are deformed or modified due to transformations such as translation, rotation and scale change, although some of them may work well even with large additive uncorrelated noise in the data.
 - When the data is directly presentable to the classification network, the direct applications are highly successful.
 - When the data is clean and ready to input then it is easy to process for multilayer neural network. Limits of classification performance will be reached if the symbols are degraded due to 'deformations', in the case of Olympic symbols or if the input corresponds to casually 'handwritten'.
 - Characters in the case of character recognition, or if the 'knowledge' of the bidding sequence and the 'reasoning' power of human players have to be used in the bridge bidding problem.

Q.2 Explain recognition of Olympic games symbols using ANN.

[SPPU : 10 Marks]

Ans. : Recognition of Olympic games symbols :

The problem - Recognition of Olympic games symbols.

The data -

- All the symbols are represented as black and white pixels on a 128×128 points grid. Although the symbols appear complex in terms of detail, each symbol represents a rigid-object-like behaviour. This behaviour ensures that the relative pixel positions of a symbol do not change even under severe degradation, such as translation, rotation and scaling.
- From above 128×128 points images set, a 16×16 element array is reconstructed as shown below.

Pattern Classification Process

1. Here, Hamming network which performs template matching using neural principles is suitable for classification. The Hamming network is a maximum likelihood classifier for binary inputs and it performs correlation matching between the input and the stored templates.
2. It consists of two subnets as shown in below Fig. Q.2.1. The lower subnet consists of $M(128 \times 128)$ input units, each corresponding to a pixel in the given pattern and N output units corresponding to the N pattern classes, which in this case is 20.
3. In the lower subnet the connection weights between the input and output units are fixed in such a way that the network calculates the distance from the input pattern to each of the N stored pattern classes. The weights are given by below equation,

$$w_{ij} = \alpha_{ij}/2, \theta_i = M/2, 1 < j \leq M, i \leq I \leq N \quad \dots(Q.2.1)$$

where w_{ij} is the connection weight from the input unit j to the output unit i in the lower subnet, θ_i is the threshold for the i^{th} output unit and a_{ij} is the element j of the pattern for the i^{th} symbol. The values of a_{ij} are - 1 or 1.

4. In the upper subnet, the weights are fixed in such a way that the output units inhibit each other. That is,

$$\begin{aligned} v_{kl} &= 1, \quad \text{for } k = l \\ &= -E, \quad \text{for } k \neq l \end{aligned} \quad \dots(Q.2.2)$$

where v_{kl} is the connection weight between the units K and l in the upper net and E is a small positive number, say $E = 0.1$.

5. When a bipolar pattern is presented for classification, the lower subnet calculates its matching score (s_i) with the stored pattern for the i^{th} class as shown below,

$$s_i = s_i(0) = f\left(\sum_j w_{ij} x_j - \theta_i\right) \quad \dots(Q.2.3)$$

where $f(\cdot)$ is the output function, x_j is the j^{th} element of the input pattern and θ_i is the threshold value for the i^{th} unit.

6. The output of the lower subnet is presented to the upper subnet, where a competitive interaction takes place among the units. The dynamics of the upper subnet is given by,

$$s_i(t+1) = f\left(s_i(t) - \epsilon \sum_{k \neq i} s_k(t)\right), \quad i = 1, 2, \dots, N \quad \dots(Q.2.4)$$

7. The competition continues until the output of only one unit remains positive and the outputs of all other units become negative. The positive unit corresponds to the class of the input pattern.
8. For the set of degraded symbols given in Fig. Q.2.1, the correct classification performance is 100 % which is highly impressive, since it is difficult even for human to identify visually the discriminating features in many of these images. When the degradation is increased by reducing the resolution using an array of 8×8 elements, the recognition performance is only 13 out of 20.

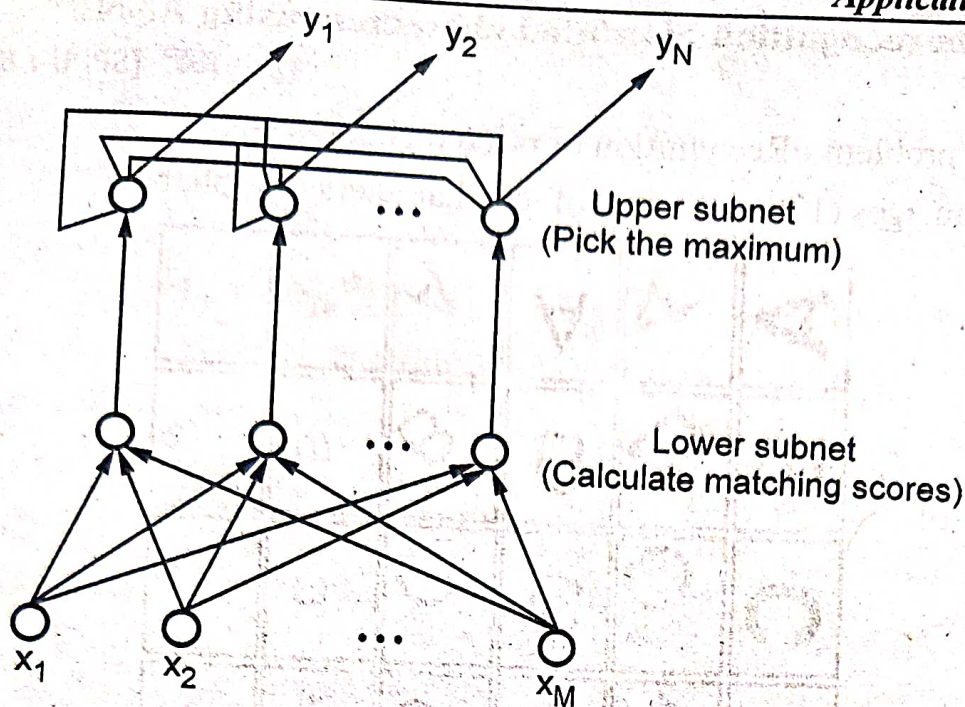


Fig. Q.2.1 The Hamming network. The input and output units are represented by x and y vectors, respectively.

9. Avoiding effects of object transformations -

(i) If the images are clean and noise-free, then there exist methods to overcome the effects of metric transformations of the objects. Several neural network models have been proposed for invariant pattern recognition.

(ii) The networks which achieve invariance by structure, the structure of the network is designed such that the output is invariant to the transformations of interest. In the case of invariance by training, representative samples of various transformations are presented during training so that the network learns equivalent transformations.

Invariance by structure or by training assumes the existence of a fixed set of weights which provide invariance over the continuum of transformations.

(iii) It also assumes that a network can be trained to estimate this set of weights from examples. But invariances cannot be built as static functions in the structure. They have to be dynamically estimated from the data. Alternatively, one can first address the transformation invariance by feature extraction and then use these features as input to a classifier.

Q.3 Explain recognition of printed characters using ANN.

[SPPU : 8 Marks]

Ans. : The problem - Recognition of printed characters.The data - images (128×128 pts) of ten characters of alphabet.

Fig. Q.3.1 Rotated, scaled and translated images of characters used for translation invariant recognition

Pattern recognition process

1. The ten characters and some transformed versions of these characters used in the current pattern recognition are shown in Fig. Q.3.1. In this case 100 % classification accuracies could be obtained for all the test data

upto a scale reduction of $1/12$ of the linear dimension of the original, that is, the reduced image is about 10×10 points.

2. Here the feature approach gives better transformation invariant recognition than in the case of the Olympic games symbols, since the objects are simpler in detail in the case of the printed characters of the alphabet.
3. The pattern classifications can be accomplished using neural network models for objects whose images are severely degraded by transformations and noise. But in all these cases the objects considered are assumed to be rigid, in the sense that there was no relative displacement in different parts of the object.
4. It should be noted that the above illustrations differ from the handwritten characters in which different parts of a character are deformed differently in each sample.
5. Thus the classification methods based on correlation matching or training a feedforward network using moment features are not useful for problems such as recognition of handwritten characters.

Q.4 Explain Neocognitron application used for hand written character recognition.

 [SPPU : 8 Marks]

Ans. :

1. The neocognitron is a **hierarchical multilayered neural network** proposed by Professor Kunihiko Fukushima for **handwritten character recognition**.
2. There are various different versions of the neocognitron. Two original basic versions proposed by Professor Fukushima differ in used **learning principle** namely,
 - Learning without a teacher
 - Learning with a teacher.
3. The first version of the neocognitron was based on the **learning without a teacher**. This version is often called **self-organized neocognitron**.

4. The **main advantage** of neocognitron is its ability to recognize correctly not only learned patterns but also patterns which are produced from them by using of partial shift, rotation or another type of distortion.
5. **Feature extraction in neocognitron - Hierarchical feature extraction** is the basic principle of the neocognitron. Hierarchical feature extraction consists in **distribution of extracted features to several stages**. The simplest features (usually only rotated lines) are extracted in the first stage and in each of the following stages the more complex features are extracted. In this process it is important fact that only informations obtained in the previous stage are used for feature extraction in the certain stage.

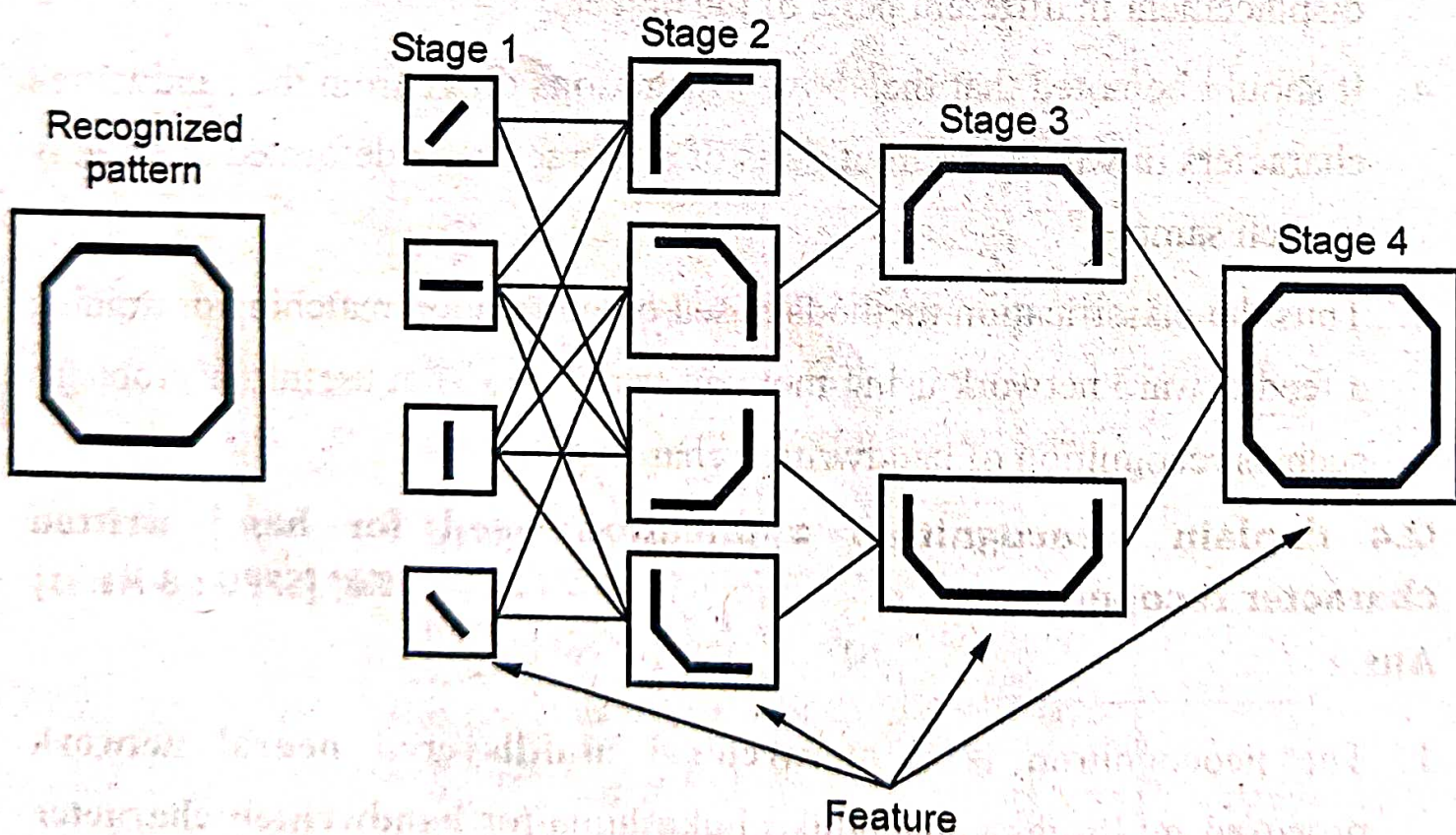


Fig. Q.4.1 Hierarchical feature extraction

6. Neocognitron architecture

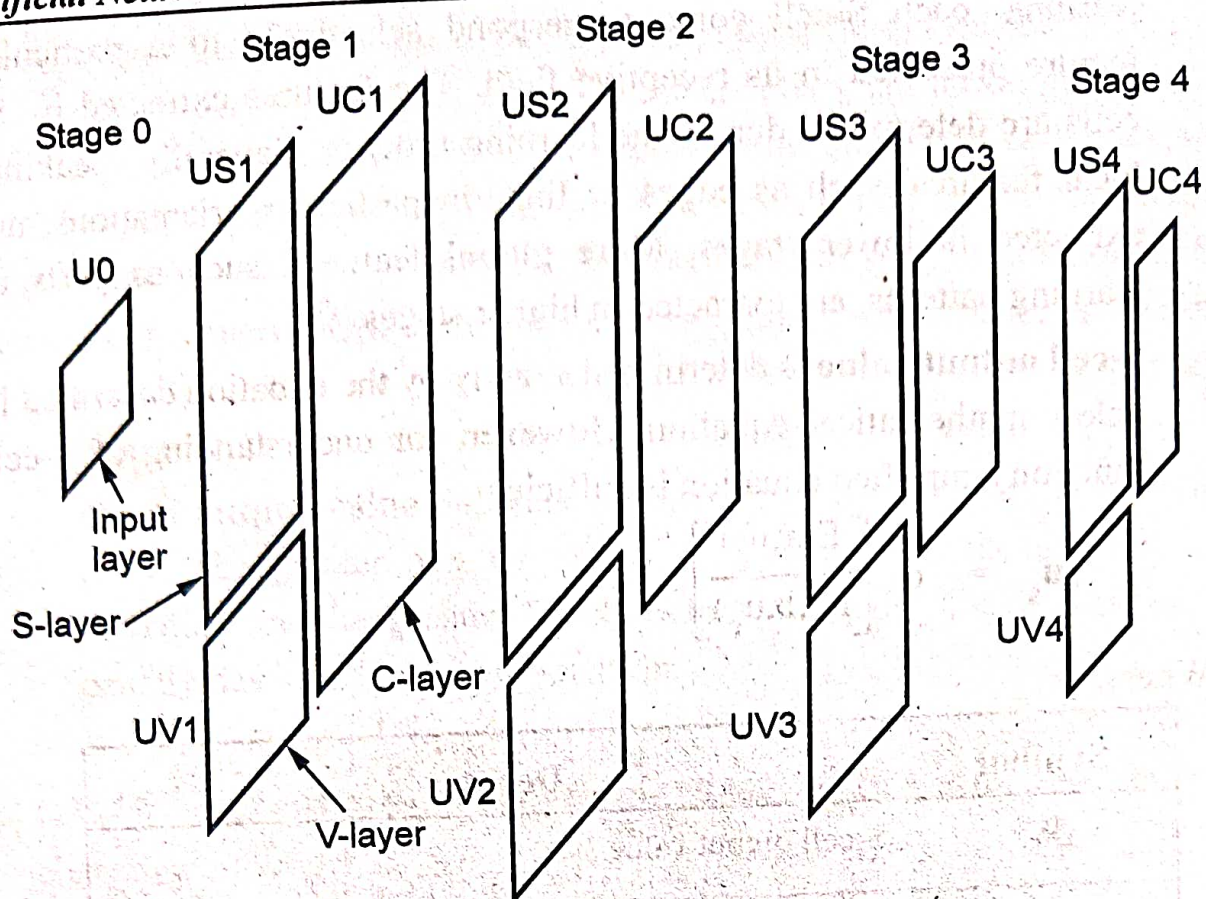


Fig. Q.4.2 Neocognitron network structure

Symbol	Denotes
U0	Input layer
USl	S-layer in the l-th stage of the network
UVl	V-layer in the l-th stage of the network
UCl	C-layer in the l-th stage of the network

- i) A typical architecture of the network has the lowest stage which is the input layer consisting of two-dimensional array of cells, which correspond to photoreceptors of the retina. There are retinotopically ordered connections between cells of adjoining layers. Each cell receives input connections that lead from cells situated in a limited area on the preceding layer. Layers of "S-cells" and "C-cells" are arranged alternately in the hierarchical network. In the above figure the contrast-extracting layer is inserted between the input layer and the S-cell layer of the first stage.
- ii) **S-cells** work as feature-extracting cells. They resemble simple cells of the primary visual cortex in their response. Their input connections are variable and are modified through learning. After having finished

6.2 : Applications of ANN in Speech Recognition

Q.5 How ANN can be used for English text conversion to speech application?

OR Write a note on NETTalk application.

[SPPU : 10 Marks]

Ans. :

1. **Speech Signal** - Speech signal is the output of a time - varying vocal tract system excited by a time-varying excitation signal. The vocal tract system, including the coupling of the nasal tract, can be accurately described in terms of the positions of the articulators such as tongue, lips, jaw, velum, etc.
2. Usually, the vocal tract system is approximately described in terms of the acoustic features such as the frequency response or the resonances (formants) and anti - resonances (anti-formants) of the system. These features are easier to extract from the signal than the articulatory parameters.
3. The excitation of the vocal tract system consists of broadly three categories,
 - i) Voiced source (the quasiperiodic excitation due to the vibrating vocal folds).
 - ii) Unvoiced source (the turbulent flow of air at a narrow constriction created in the vocal tract during production)
 - iii) Plosive source (the abrupt release of the pressure built up behind a closure in the vocal tract system). The voiced source is characterized by the periodicity (pitch period) and the change of the pitch period with time (intonation).
4. The short -time characteristics of the speech signal are represented by the short- time (10-20 ms) spectral features of the vocal tract system as well as the nature of excitation in the short-time segment. These are called **segmental features**. Supra- segmental features of speech are represented by the variation of the pitch period (intonation), the durations of different sound units, and the coarticulation reflecting the dependence of

characteristics of one sound unit on the neighbouring sound units during speech production.

5. Speech is a sequence of sound units corresponding to a linguistic message. Below are important applications in speech area,

- a) **Speech recognition** - The objective is to determine the sequence of sound units from the speech signal so that the linguistic message in the form of text can be decoded from the speech signal.
- b) **Speech synthesis** - The objective is to determine the sequence of sound units corresponding to a text so that the given text message can be encoded into a speech signal.
- c) **Speaker identification** - The objective is to determine the identity of the speaker from the speech signal.

6. Problems in speech recognition problems -

- i) The main challenge and problem in the speech applications is processing of the speech signal in a manner similar to human auditory processing mechanism, so that features relevant to a particular task can be extracted.
- ii) The speech problem is further complicated by the fact that the message is conveyed not only through the segmental features but also by the suprasegmental features. It is the lack of understanding of these segmental and suprasegmental features and their extraction mechanism that makes the speech tasks extremely difficult for implementation by machines.

7. The NETtalk working

- (i) It is a multilayer feedforward neural network (Fig. Q.5.1) developed to generate pronunciation units or phoneme code from an input text. The phoneme code is presented as input to a speech synthesizer to produce speech corresponding to the text.
- ii) The network consists of an input layer of $7 * 29$ binary units, corresponding to 7 input text characters including punctuation, one hidden layer with 80 units and one output layer with 26 units corresponding to 26 different phonetic units.

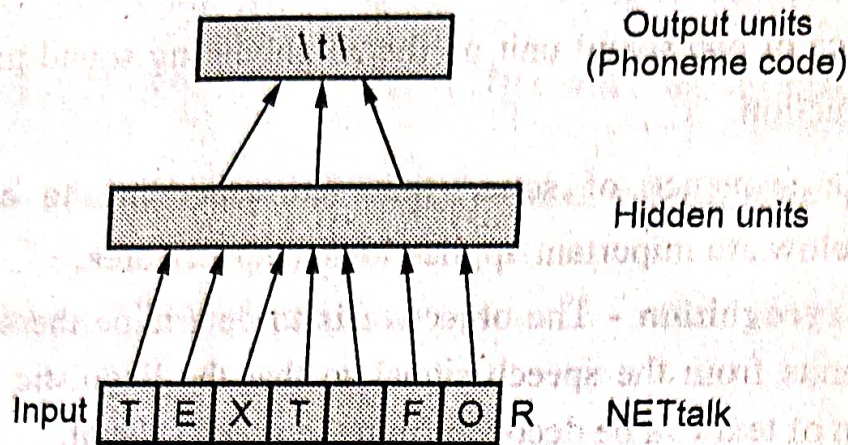


Fig. Q.5.1 NETtalk - feedforward network to convert English text to speech

- iii) The network takes as input 7 consecutive characters at a time and produces the phoneme corresponding to the pronunciation of the letter at the centre of the input string. Each character of the input is represented as a 29-bit string. The network was trained on 1024 words from a set of English phoneme exemplars giving the pronunciation of words, and was tested using different text sentences. The system produces a string of phonemes for a given input text. The speech produced by a synthesizer using these strings of phonemes as input was intelligible, although the quality was poor.
- iv) The network was capable of distinguishing between vowels and consonants. On a new text the network achieved a generalization accuracy of 78 % in phonetic transcription. The system merely demonstrated the ability of a neural network to capture the letter to sound rules from input-output data. The quality will be obviously poor, as it cannot capture the significant suprasegmental features which are essential for producing natural sounding synthetic speech.

Q.6 Brief on vowel and consonants segment recognition, texture classification.

OR How ANN is used in recognition of consonants, vowels, texture classification and segmentation ?

Ans. : [SPPU : 8 Marks]

1. Consonant-Vowel (CV) utterance typically forms a production unit (syllable) in speech and hence several attempts have been reported for recognition of CV utterances. Since these are dynamic sounds, the spectral patterns change with time. Each utterance of a CV unit is

represented as a temporal sequence of spectral vectors. Each spectral vector corresponding to a fixed 10 ms segment may be represented using 16 log spectral coefficients on a mel-frequency scale or using the corresponding mel-scale cepstral coefficients.

2. The number of spectral vectors per CV utterance generally varies. But usually a fixed duration (50-200 ms) segment of CV enclosing the vowel onset, the transition to vowel and some steady part of the vowel, is used to represent a CV unit. The CV units are thus temporal sequence patterns and hence static pattern recognition networks like Multilayer Feedforward Neural Network (MLFFNN) are not suitable for recognition of these units.
3. Moreover, discriminability among these CV units is low due to domination of the vowel context. Straight forward method to perform sequence recognition is to view the temporal sequence of the spectral vectors as a two-dimensional spatial input pattern for a MLFTNN. The conventional backpropagation learning can then be used to train the network. A better approach for CV recognition is through Time Delay Neural Networks (TDNN). TDNN is a MLFFNN with its input consisting of time-delayed input frames of data. The input to the intermediate hidden layers also consists of time-delayed outputs of the preceding layer. Below figure illustrates the idea of a time-delay,

(See Fig. Q.6.1 on next page)

In the Fig. Q.6.1 the neural network is applied for classification of three CV units lbl, Id and Igl. The time sequence is reflected in, the computation of block of three frames of data at a time, with two frames overlapping for successive blocks. The Fig. Q.6.1 shows multiple copies of the TDNN aligned with adjacent input spectral vectors. The first TDNN is shown in the boxes marked by thick lines. In this case each utterance has 15 frames of data, each frame consisting of 16 spectral components. For this, 13 different copies of TDNN are created. These include 13 copies of the input layer, nine copies of the first hidden layer,

and only one second hidden layer and one output layer. The output layer has three units corresponding to the three classes Ibl, ldl, and Igl. For each TDNN, each unit in a layer is connected to all the units in the layer below it. For each TDNN there are 16×3 units in the input layer, 8×3 units in the first hidden layer and 9×3 units in the second hidden layer and 3 units in the output layer. Multiple copies of the TDNN as shown in the Fig. Q.6.1 enable the entire history of the network activity to be present at the same time. This allows the use of the backpropagation learning algorithm to train the network.

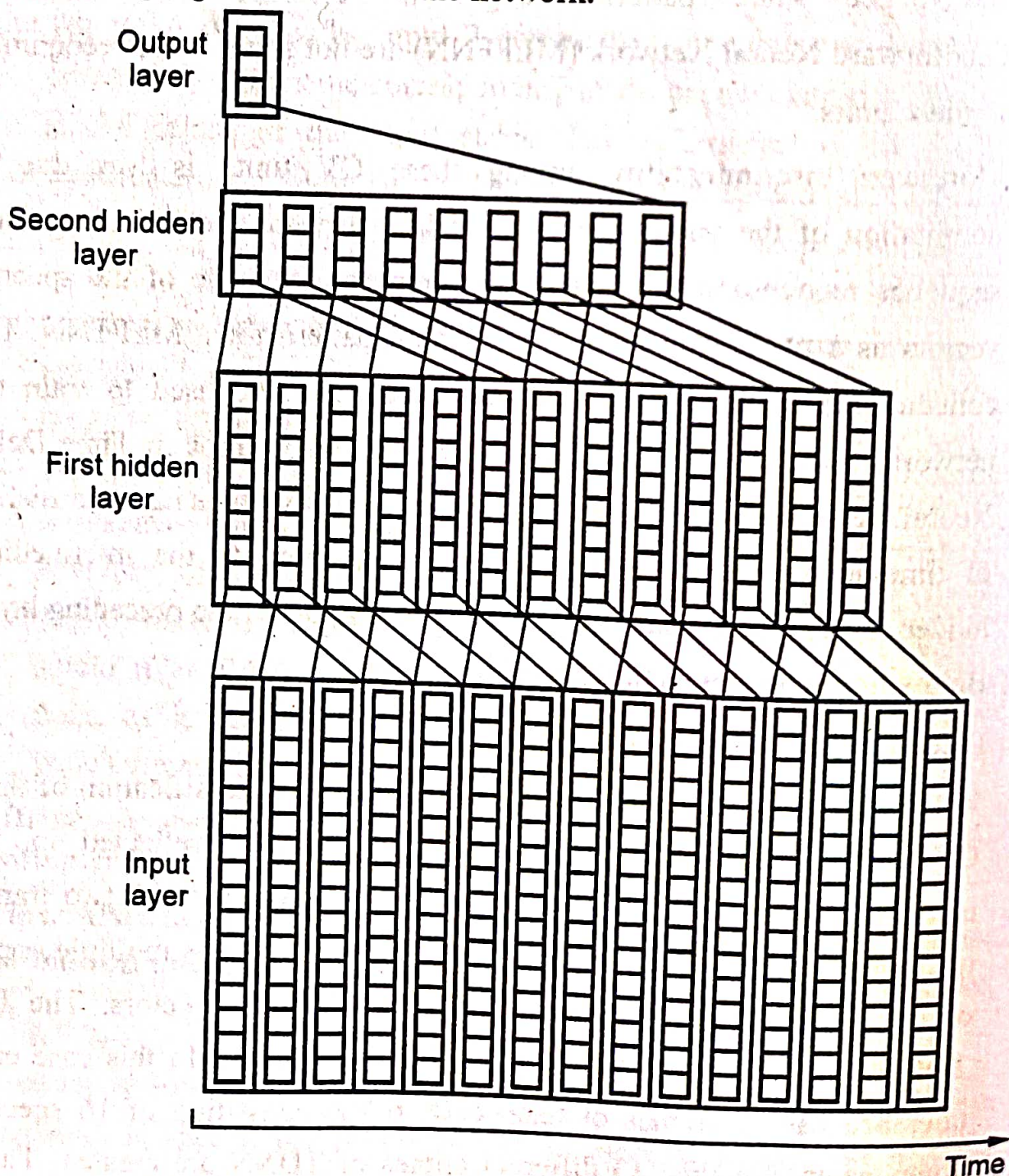


Fig. Q.6.1 Time delay neural network architecture

- Types of transfer learning are Domain Adaptation, Domain Confusion, Multi-task Learning, One-shot Learning and Zero-shot Learning.
Also refer Q.32 of Chapter - 5.

Q.7 a) Which device recognize a pattern of handwritten or printed characters ? And also illustrate it's working. (Refer Q.3 of Chapter - 6) [7]

b) Explain texture classification using convolution neural network. [5]

Ans. : Texture classification is a fundamental issue in computer vision and image processing that plays a significant role in many applications such as medical image analysis, remote sensing, object recognition, document analysis, environment modeling, content-based image retrieval and many more.

- Texture classification is a complex task that assigns unlabeled images or textures to the correct labeled class. Some key factors such as scaling and viewpoint variations and illumination changes make this task challenging.
- Almost all of the CNN model cases classify images based on global features that describe the image as a whole to generalize the entire object.
- The primary concept of using CNN features is its capability to capture the high-level features that are considered the key point to separate the texture image. Convolution Neural Network (CNN) captures low detail features from the first convolution layers and then generates high detail features in the last layers. Selecting the proper right set of features has always been a challenging task for the machine learning algorithm. CNN has numerous extensive applications in different fields. Whereas image classification and recognition are challenging tasks for machine learning algorithms, CNN is highly promising for accuracy.

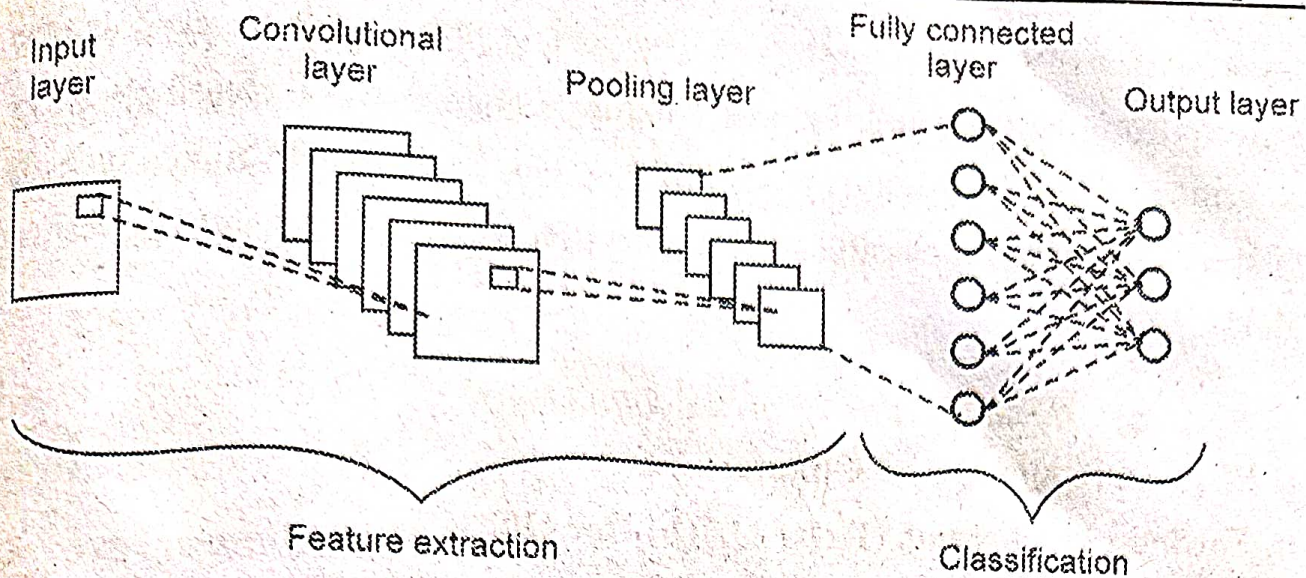


Fig. 1

As shown in above Fig. 1 feature extraction and classification involves different layers of network as discussed below.

- **Convolution layer :** It is used to apply filters to the input feature map and produce convolved features. The following two factors define the convolution layer. (1) The size of the block extracted from the input feature map. (2) The depth of convolved features.
- **Pooling layer :** Generally, applied after the convolution layer, it is a downsampling (Dimensional reduction) operation such as average pooling, min pooling and max pooling.
- **Fully connected layer :** It is typically found towards the end of the CNN architecture. It works on such a flattened input, in which each input is linked to all nodes.

Each model of CNN has its characteristics. Fig. Q.5.3 of Chapter - 5 summarizes the description of different CNN arch.

The motivation of the usage of CNN models as features descriptors is the ability of the deep neural network to capture the high-level features that can be a key point for the classification of the texture images. Texture plays a significant role in distinguishing objects in color images. Texture classification involves a two-phase process. The first phase is extracting the features, which provides a feature-based description for each texture type; this phase tends to select features unaffected by image transformation, such

as scaling, translation and rotation. The second phase tends to recognize the texture from the extracted features. Texture recognition is increasingly set as a critical issue in computer vision that has many applications such as biomedical image processing, object detection and remote sensing application fields.

c) Write short notes on the following :

- i) NET talk (Refer Q.5 of Chapter - 6)
- ii) Texture classification (Refer Q.7(b) Dec.-2023)
- iii) Pattern classification (Refer Q.1 of Chapter - 6).

[5]

OR

Q.8 a) You have been asked to develop a model of recognizing hand written digits. What are the chosen steps for activity ? Explain each with detail. (Refer Q.3 of Chapter - 6)

[7]

b) What is automatic translation ? How does it work ? What are its benefits ?

[5]

Ans. : Automatic translation refers to the process of replacing words from one language with their counterparts in another language. It involves the use of software that can **rapidly translate text** with minimal human intervention.

- Typically, automatic translation relies on corpora, which are collections of texts in multiple languages. Words are translated automatically by analyzing their context and proximity to other words. However, due to the inherent limitations of machines, automatic translation cannot provide a **flawless translation**. It lacks the ability to fully comprehend context and nuanced meanings.
- Neural Machine Translation (NMT) is typically software used to translate words from one language to another. Google Translate, Baidu Translate are well-known examples of NMT offered to the public via the Internet.
- The reason this NMT is important is because recent advancements in the technology have allowed an increasing number of multinational institutions to adopt NMT engines to aid in internal and external communications.

- Technically, NMTs encompass all types of machine translation where an artificial neural network is used to predict a sequence of numbers when provided with a sequence of numbers. In the case of translation, each word in the input sentence (e.g. English) is encoded as a number to be translated by the neural network into a resulting sequence of numbers representing the translated target sentence (e.g. Chinese).

ANN benefits :

- Neural networks can have a large number of parameters with weights and biases between nodes to give them the flexibility to fit highly complex data and train complex models. Model complexity allows the model to generalize to the large volumes of examples it's trained on, such as digesting millions of language pairs.
- One way to understand neural networks is to think of the input as a signal with "information" in it. Some of this information is relevant for the task at hand (e.g. predicting the output). Think of the output as also a signal with a certain amount of "information". The neural network attempts to "repeatedly refine" and compress the input signal's information to match the desired output signal. Think of each hidden layer of the network as stripping out unimportant parts of the input information and keeping and/or transforming the output information along the way through the network. The fully-connected neural network will transform the input information into a form in the final hidden layer, such that it is separable by the output layer.

c) What is neocognitron neural network and how it is trained ?

(Refer Q.6 of Chapter - 6)

[5]

Explaining Texture Classification

Texture : Texture analysis which consists of texture classification, segmentation and synthesis has a wide variety of applications in image processing, e.g., industrial and biomedical surface inspection, ground classification and segmentation of satellite imagery and content-based image retrieval. Texture, even though humans can effortlessly recognize texture, is very difficult to define. Classification refers to as assigning a physical object or incident into one of a set of predefined categories. In texture classification the goal is to assign an unknown sample image to one of a set of known **texture classes**. Texture classification is one of the four problem domains in the field of texture analysis. The other three are texture segmentation (partitioning of an image into regions which have homogeneous properties with respect to texture; supervised texture segmentation with a priori knowledge of textures to be separated simplifies to texture classification), texture synthesis (the goal is to build a model of image texture, which can then be used for generating the texture) and shape from texture (a 2D image is considered to be a projection of a 3D scene and apparent texture distortions in the 2D image are used to estimate surface orientations in the 3D scene). Texture analysis is important in many applications of computer image analysis for classification or segmentation of images based on local spatial variations of intensity or color. A successful classification or segmentation requires an efficient description of image texture. Important applications include industrial and biomedical surface inspection, for example for defects and disease, ground classification and segmentation of satellite or aerial imagery, segmentation of textured regions in document analysis and content-based access to image databases. However, despite many potential areas of application for texture analysis in industry there is only a limited number of successful examples. A major problem is that textures in the real world are often not uniform, due to changes in orientation, scale or other visual appearance. In addition, the degree of computational complexity of many of the proposed texture measures is very high.

c) Discuss the application of ANN in the recognition of Consonant Vowel (CV) segments. (Refer Q.6 of Chapter - 6)



